

Data Anonymisation Rules

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Anonymisation

The Anonymisation class is to contains a list of methods that can be used to consistently apply anonymisation rules to sensitive data. Each method will utilise, where appropriate, the faker library installed on the Databricks cluster. The methods described below are to be used for all anonymisation scenarios which are to be implemented in Databricks notebooks and supporting patterns. This includes (but not limited to) the following scenarios/types:

Anonymised to reflect deletion at source

- Anonymisation of Data deleted at source

Anonymised without losing business value

- Anonymisation of source data where the source system does not have anonymised data (dev. env. only)
- Anonymisation of data for successful DSAR requests.
- Anonymisation of data copied to the open lake.

Anonymisation Methods

The following define the requirements of each method to be implemented:

AnonymisePhoneNumber

- Expects a string as a parameter
- Replaces phone number with a random UK style phone number using faker library

Before	After
01480 123456	01743 654321

AnonymiseMobileNumber

- Expects a string as a parameter
- Replaces mobile phone number with a random UK style mobile phone number using faker library

Before	After
+447500111111	+447501222222

AnonymiseFullName

- Expects a string as a parameter (first and second names concatenated)
- Replaces string with random first name and random second name separated by a single space.

Before	After
Mick Morgan	Rory Smith

AnonymiseLogon

- Expects a string as a parameter (first and second names concatenated)
- Replaces string with first letter of a random first name concatenated to a random second name all in upper case.

Before	After
Mick Morgan	RSMITH

AnonymiseFirstName

- Expects a string as a parameter
- Replaces string with a random first name.

Before	After
Mick	Rory

AnonymiseLastName

- Expects a string as a parameter
- Replaces string with a random last name.

Before	After
Morgan	Smith

AnonymisePostalCode

- Expects a string as a parameter and an integer mask length.
- If segment length >0 it will replace postcode with a partially masked value of the original postcode defined by the segment length. If segment length is 0 replace postcode with random uk style postcode.

Segment value	Before	After
2	PE29 7DY	PE
4	PE29 7DY	PE29
6	PE29 7DY	PE29 7
0	PE29 7DY	PE99 9XX

AnonymiseBuildingNumber

- Expects a string as a parameter.

- Replaces the building number with random UK building number..

Before	After
22	67

AnonymiseStreetAddress

- Expects a string as a parameter.
- Replaces the address with random UK street address.

Before	After
Studio 17B	Beuna Vista

AnonymiseStreetName

- Expects a string as a parameter.
- Replaces the address with random UK street name.

Before	After
Lancaster Way	Miller Drive

AnonymiseCity

- Expects a string as a parameter.
- Replaces the city with random postal town/city.

Before	After
Huntingdon	Lincoln

AnonymiseEmail

- Expects a string as a parameter
- Replaces email with a random ascii safe email address.

Before	After
peter.rabbit@anglianwater.co.uk	bartlettjulia@example.com

AnonymiseDate

- Expects a string as a parameter
- Replaces the day and month with 01 and returns the year before. All dates will be formatted with the character /.

Before	After
01/08/2018	01/01/2017

AnonymiseGender

- Expects a string as a parameter, although it is not used in the method. It is included in case there's a need to change the anonymisation rule using the string.
- Replaces the gender with the character U.

Before	After
Male	U

AnonymiseMaritalStatus

- Expects a string as a parameter, although it is not used in the method. It is included in case there's a need to change the anonymisation rule using the string.
- Replaces the marital status with the text Unknown.

Before	After
Single	Unknown

AnonymiseNationality

- Expects a string as a parameter, although it is not used in the method. It is included in case there's a need to change the anonymisation rule using the string.
- Replaces the nationality with the text Unknown.

Before	After
British	Unknown

IntFromString

- Expects a string as a parameter, returns int. It does not matter if it is an empty string, the return will always be a numerical value, defaults to 0,
- Supporting method for above name functions that require an integer to be passed in as the first parameter where an integer may not be available, gets last character and returns character position.

Before	After
jSmith	8
jSmith4	4
jSmith^\$)	0
jSmithR	18

AnonymiseVehicleRegistration

- Expects an integer and a string as parameters, although only the integer is used in the method. The string is included in case there's a need to change the anonymisation rule using the string.
- The method invokes the class UserTable to create a configuration table as shown below:

Vehicle Id (Natural Key)	Vehicle Registration
0	AA00AND
1	BB11BAR
2	CC22CAT
3	DD33DOG
4	EE44EGG
5	FF55FLY
6	GG66GAP
7	HH77HAT
8	II88ICE
9	JJ99JAR

- The method extracts the last digit of the vehicle ID (integer) and returns the anonymised from the configuration table

Vehicle ID (Natural Key)	Vehicle Registration Before	Vehicle Registration After (Anonymised)
1234	<Actual Vehicle Reg.>	EE44EGG
4321	<Actual Vehicle Reg.>	BB11BAR